Control Major Bleeding

Consider coordinating with rescue personnel on release timing

Administer Oxygen
To maintain Sp02 >94%

EMR

Consider Tourniquet Application

EMT

Note Time of Tourniquet Application on Tourniquet

Establish large bore IV access

EMT

Where possible, start the administration of fluids prior to release

Consider IV NaCl (0.9%)
Adults: 500mL
Pediatric: 20mL/kg

EMT

Go to Pain CPG

Pain Management

Go to Trauma Management CPG

Standard Trauma Care during and post extrication

12-Lead ECG and continuous 4 lead ECG monitoring

Patient Care Goals

1. Recognize and manage systemic complications of crush injuries early
2. Prevent or minimize the risk of crush syndrome, hyperkalemia, and acute kidney injury
3. Relieve compressive forces safely and in a controlled manner
4. Provide adequate pain relief and support oxygenation and perfusion
5. Coordinate safe extrication, fluid resuscitation, and timely hospital transport

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patient entrapped or compressed by an object for >30 minutes
- Clinical evidence of extremity or torso crush injury
- Suspected or confirmed muscle injury with systemic symptoms
- Scene or dispatch information suggesting prolonged compression or entrapment

Exclusion Criteria

- Minor soft tissue contusions without systemic risk
- Isolated fractures without prolonged entrapment
- Patients with fatal injuries not compatible with life

Key Patient Assessment Points

1. **Initial Impression**
   - Scene safety, mechanism of injury, time under compression
   - Number of patients, need for extrication or heavy rescue
2. **Airway & Breathing**
   - Assess for trauma-related airway compromise, hypoxia, or impaired ventilation
   - Administer high-flow oxygen if SpO_2 <94% or chest trauma present
3. **Circulation & Perfusion**
   - Check for tachycardia, hypotension, or signs of shock
   - Inspect compressed limb for swelling, pallor, cyanosis, coolness
   - **Capillary refill, peripheral pulses, and distal sensation**
4. **Crush Syndrome Warning Signs**
   - Prolonged entrapment (>30 minutes)
   - Dark or decreased urine output (if catheter or history known)
   - Swelling and firmness of limb
   - Muscle weakness, paresthesia, or paralysis
5. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority: Entrapment >30 mins, systemic symptoms, hypotension, altered mental status
     - Yellow Priority: Shorter duration entrapment, isolated limb swelling, stable vitals
     - Green Priority: No systemic signs, minor soft tissue crush injury

Patient Management & Interventions

1. **Consider ALS Backup**
   - ALS back-up should be routinely requested for ALS:
     - Prolonged entrapment (>30 min)
     - Suspected crush syndrome
     - Cardiac dysrhythmias
     - ALS medications (e.g. pain relief)
2. **Pain Management**
   - Assess and manage pain as per Pain Management CPG.
3. **Oxygen Therapy**
   - Administer high-flow oxygen (target SpO_2 ≥94%)

- Continue oxygen throughout extrication and transport

4. **IV Access & Fluid Resuscitation**
   - Establish large-bore IV access before extrication if entrapment >30 mins
   - Administer initial bolus of Normal Saline, repeat as needed to maintain perfusion.
     - Adults: 500mL
     - Pediatric: 20mL/kg
   - Anticipate hyperkalemia, acidosis, and rhabdomyolysis once compression is released
5. **Tourniquet Application**
   - If blood flow to extremity is impaired and extrication will release toxins into circulation, consider tourniquet placement prior to release.
6. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - ECG Monitoring:
       - Continuous ECG monitoring (watch for peaked T-waves, wide QRS)
     - Compartment Syndrome:
       - Monitor for deterioration or signs of compartment syndrome
7. **Transport**
   - Notify receiving facility early if prolonged entrapment or suspected crush syndrome
   - Transport urgently with continued fluids, oxygen, and monitoring

Patient Safety Considerations

- Do not delay IV fluids until after release — pre-hydration improves outcomes
- Use Broselow tape for pediatric patients to guide equipment and dosing
- Avoid overuse of opioids that may mask declining LOC or respiratory effort
- Apply tourniquet judiciously and only if limb injury is severe and extrication imminent
- Be prepared for sudden deterioration post-release

Documentation

- Identification: Age, gender, weight
- Medical complaint: Crush injury and entrapment time
- Injuries/Information: Affected area, limb deformity, swelling, neurovascular status
- Signs: Vitals, perfusion, ECG if applicable
- Treatment: Oxygen, fluids, pain relief, tourniquet
- Allergies: If known
- Medications: Prehospital analgesia, IV fluids, ALS medications
- Background: History of renal issues, previous episodes
- Other: Entrapment duration, scene challenges, time of release

Clinical Insights & Additional Guidance

- Crush syndrome can lead to fatal arrhythmias, acute kidney injury, or shock; prevention starts in the field
- Fluid before release is one of the most critical prehospital interventions
- Be alert for signs of hyperkalemia, especially in crush injuries involving large muscle groups or prolonged entrapment. Clinical signs may include:
  - Muscle weakness or flaccid paralysis
  - Nausea, vomiting, or paresthesia
  - Bradycardia, hypotension, or irregular pulse
  - ECG may show peaked T-waves or widened QRS — but may appear normal early
  - Treat based on clinical suspicion, not ECG alone. Early fluid resuscitation is critical.
- Delays in ALS arrival must be mitigated by early BLS actions
- Tourniquets should not be used for routine crush injuries unless release is imminent and limb is non-viable

- Electrical crush injuries (e.g., pinned under live machinery) require additional precautions
- When in doubt, contact Tele-EMS and prepare early hospital notification

Key Performance Indicators

1. Document entrapment duration of ePCR.

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